

AttentionGo:

Designing Tangible Neurofeedback Tools for  
Children with ADHD through a Holistic Well-being

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Allow children to  
personalize and  
decorate the game



Allow children to use  
their own belongings as  
elements of play



Encouraging  
children to explore  
an open-ended map

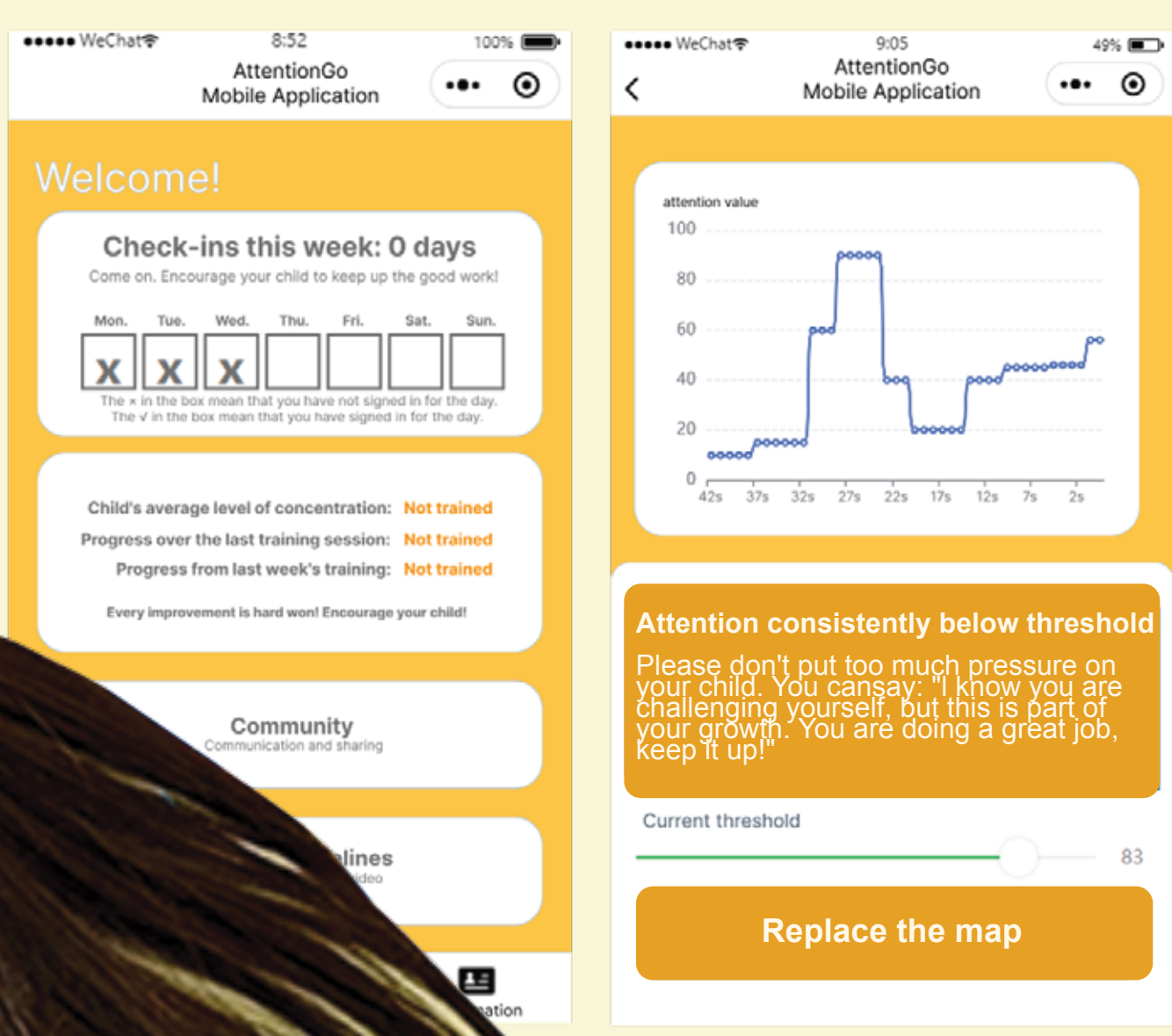


Supports for  
co-play to  
promote  
social  
connection



Framing perceived progress by  
designing growth-oriented game  
structures

Providing  
real-time  
guidance  
for adults  
stakeholder



Presenting  
historical and  
interpretive  
summaries

Tangible  
Interaction  
Design

Companion Mobile Application

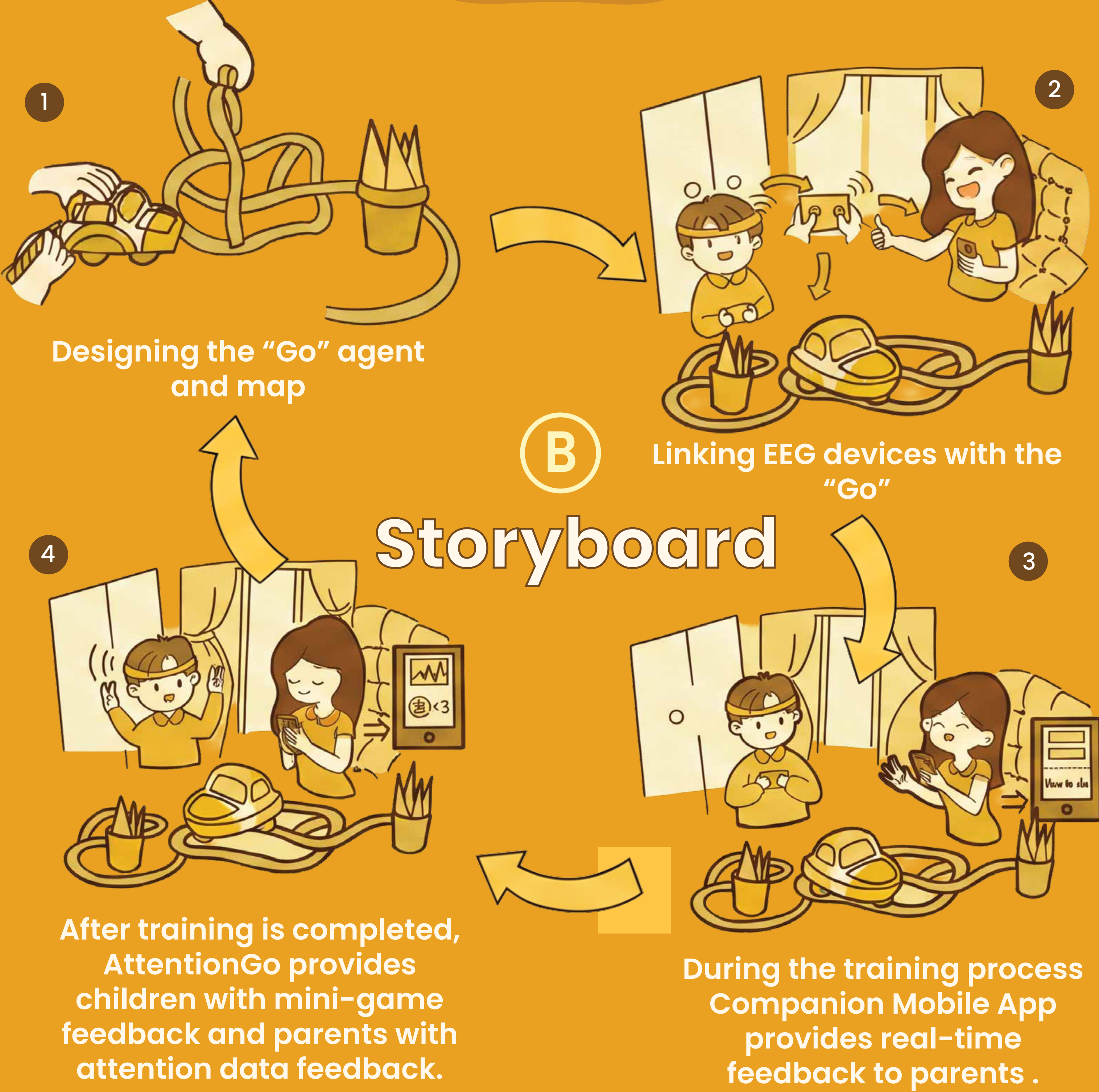


Recommended reading order: from **A** to **G**

Background

**A**

Neurofeedback training (NFT) has gained traction as a non-pharmacological intervention for children with ADHD, yet most existing systems are designed from a neurotypical perspective that neglects the well-being of neurodivergent users. To address this gap, this paper presents AttentionGo, a tangible neurofeedback system grounded in Self-Determination Theory (SDT), which informs its core design strategies.



Design Requirements

**C**

- 1 Encourage creative, self-directed, and ludic play experiences.
- 2 Enable game rules that make progress visible and meaningful to children.
- 3 Support meaningful stakeholder participation to foster social connectedness.
- 4 Address usability issues.

AttentionGo System

**D**



1

GO Tangible Agent

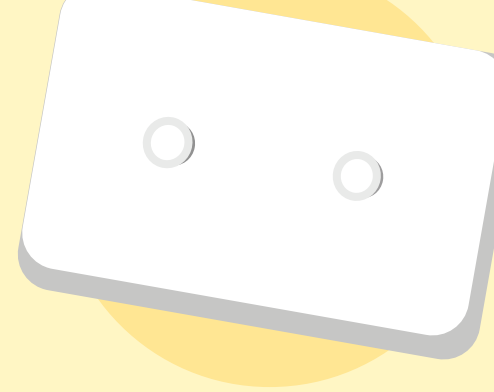
"GO" Tangible Agent is a Raspberry Pi-powered robotic vehicle whose speed dynamically reflects the child's real-time attention level during NFT.



2

EEG Headband

EEG Headband captures the child's brain signals and maps attention scores (0-100) to GO's forward speed, transmitting data wirelessly via Wi-Fi.



3

Handheld Controller

Handheld Controller is a ESP-32-based device with two buttons and voice guidance that manages basic operations.



4

Companion Mobile Application

Companion Mobile Application is a caregiver-facing interface that delivers real-time supportive prompts and visualizes session summaries.



5

Three Growth-themed Mini-Games

Three Growth-themed Mini-Games provide children with stage-based visual rewards that evolve across training sessions.

Applying SDT to NFT

**E**

Autonomy

AttentionGo fosters autonomy by allowing children to freely customize the controller, GO shell, and game map. However, future designs should carefully balance creative freedom with training effectiveness, as open-ended play may lead to prolonged sessions.

Competence

AttentionGo supports competence by dynamically adjusting attention thresholds and incorporating growth-oriented mini-games. However, careful calibration of task difficulty is essential, as poorly tuned challenges may undermine children's sustained engagement in NFT.

Relatedness

AttentionGo fosters relatedness through parental involvement via the companion app, therapist data sharing, and a co-play mode. However, these features remain empirically unvalidated and raise concerns around privacy and performance pressure that warrant further investigation.

Limitations

**F**

- No empirical evaluation yet
- The lack of a formal co-design process or pilot study involving children with ADHD
- The signal noise inherent in consumer-grade EEG headbands

Future Work and Conclusion

**G**

AttentionGo has not yet been empirically evaluated; future work will involve participatory workshops, longitudinal home deployment, and feasibility studies, with open questions remaining around data transparency and broader applicability to neurodiverse populations.